

A three-prime exclusion for the quadratic-residue divisibility criterion

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<https://rsihouse.ai/openmath> ; <https://luma.com/yzp9abvr> ;

<https://www.sundai.club/events/boston/recursive-learning-hack-with-harvard-innovation-labs>

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Central source and adjudication archive: <https://github.com/alejandrozu/openmath-2026-judging>

Abstract

For $n=pqr$ with three distinct odd primes, the quadratic-residue count $a(n)$, including zero, satisfies $a(n)(a(n)-1)$ not dividing n^2-1 . A global arithmetic reduction bounds the first two primes and the quotient; exact discriminant certificates exclude all remaining cases.

The residue convention and theorem

Let $a(n)$ count all square residue classes modulo n , including zero. For distinct odd primes $2 < p < q < r$, the Chinese remainder theorem gives $a(pqr) = (p+1)(q+1)(r+1)/8$. The theorem excludes divisibility of n^2-1 by $a(n)(a(n)-1)$ throughout this unbounded squarefree three-prime family. It therefore establishes the corresponding conjectured equivalence on this entire factorization class.

Including zero is essential: a count of only nonzero quadratic residues would change the arithmetic. The theorem says nothing about repeated prime powers or products of four or more distinct primes, and should be described as a uniform infinite subcase of A000224 rather than the full conjecture.

Global reduction to a finite certificate

Set $A=(p+1)/2$, $B=(q+1)/2$ and $C=(r+1)/2$. Under a contrary divisibility assumption, there is a positive integer quotient K satisfying $(pqr)^2 = KABC(ABC-1)+1$. The proof derives necessary bounds from that exact equation, after explicitly justifying every conversion from natural-number subtraction to ring subtraction.

The key global step bounds p by 381 and gives a p -dependent upper bound on q . With p and q fixed, K lies in an explicit finite interval. The remaining equation is quadratic in C . Writing $H=pq$ and $U=AB$, its coefficients are $4H^2-KU^2$, $-(4H^2-KU)$, and H^2-1 . An integral root forces the corresponding

discriminant to be an integer square.

The certificates reject that finite quotient/discriminant domain using strict gaps between consecutive squares, plus a separate congruence exclusion for the exceptional square case. Because the bounds were derived without an upper bound on r , this closes an unbounded theorem. A bounded Pell scan up to a large numerical cutoff would not by itself do so.

Formal certificate architecture

The source splits the argument into square-count transport, necessary bounds, quadratic reduction and generated certificate blocks. Its 171-module record includes each intended block and three final canonical endpoints. A separate Lean 4.33.1 replay is recorded in addition to 4.34.1; tactic-only compatibility changes in Bounds preserve definitions and theorem statements. Several early source comments still say unbuilt draft, so the dated manifest and selected endpoint receipts should govern verification descriptions.

For publication, append a compact definition of the finite domain and the discriminant identity, with machine-readable certificate data in the repository. The long generated leaves belong in supplementary material. Report the fresh common replay status separately from archived receipts and from reviewer sampling. An independent check of 2,024 small prime triples agrees with the formula and exclusion, but is not the universal proof.

Prior scope and research contribution

CRT, quotient/Pell methods and gcd obstructions are background. Earlier records already cover simpler factorization classes and a large bounded search; the claimed delta is the global reduction and exclusion for all ordered triples of distinct odd primes. The previous scan's operational n cutoff must not be confused with an unbounded theorem merely because every local survivor happens to fall below it.

This is a focused computational number-theory paper candidate if the closest pre-event project and specifications do not already entail the theorem. It should state the factorization limitation prominently and credit the baseline methods. The number of generated modules or rejected arithmetic leaves is not a count of separate discoveries.

This short manuscript records the construction and selected endpoints. The companion Original results anthology contains the extended ordinary proof exposition and its explicitly stated premises; the preserved Lean source remains the complete formal proof record.

Verification and reproducibility

Dated partial fresh replay: the 5 October 2026, 08:56:47 UTC qualification verifies 167 of 171 frozen source modules, comprising 166 granular sources and Bounds. The retained Bounds source passed under its original policy before the controller exited 120; its unfinished receipt is preserved. The subsequent source-free initial-gate failure also remains recorded. Four whole-import sources and the three original selected endpoint prints remain unrun, so the full 171-source/3-print scope is unqualified. The mathematical claim remains restricted to three ordered distinct odd primes, using the zero-inclusive quadratic-residue count; it is not the unrestricted A000224 conjecture. Qualification: Verification/A000-actual167-abnormal-controller-exit120-partial-readonly-qualification-20261005T085647463194Z.json. Historical author/internal replay is separate, and these partial checks confer no novelty, score, award or release upgrade. A later bounded dispatch assessment ended without a new Lean child or source/audit attempt. Dated resource-only summary: Verification/root-A000-whole8GiB-bounded60-public-safe-scope-20261005.json.

OeisA224.not_divisible_three_primes

```
theorem not_divisible_three_primes {p q r : ℕ}
  (hp : p.Prime) (hq : q.Prime) (hr : r.Prime)
  (hp2 : 2 < p) (hpq : p < q) (hqr : q < r) :
  ¬ a (p * q * r) * (a (p * q * r) - 1) | (p * q * r) ^ 2 - 1
```

Source: proofs/a000224/OpHack/QuadraticResidue224/Target224.lean:33.

Source SHA-256: 3b67f7494439886af3359a9e684433f49a6dcc7d10258d1d05dcc01de7886d0

OeisA224.not_modEq_three_primes

```
theorem not_modEq_three_primes {p q r : ℕ}
  (hp : p.Prime) (hq : q.Prime) (hr : r.Prime)
  (hp2 : 2 < p) (hpq : p < q) (hqr : q < r) :
  ¬ (p * q * r) ^ 2 ≡ 1 [MOD a (p * q * r) * (a (p * q * r) - 1)]
```

Source: proofs/a000224/OpHack/QuadraticResidue224/Target224.lean:43.

Source SHA-256: 3b67f7494439886af3359a9e684433f49a6dcc7d10258d1d05dcc01de7886d0

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